



Introduction to React

1. Overview
2. Installation and First App
3. React JSX

- React is a front-end library developed and maintained by Facebook.
- It is JavaScript library for building user interfaces.
- It is component-based front end library
- React Features:
 - JSX
 - Components
 - One-way Data Binding
 - Virtual DOM
 - Simplicity
 - Performance
 - Easy learning

- Other Similar technologies available
 - **AngularJS:**
 - i. open-source JavaScript framework
 - ii. Angular arrived much earlier in comparison to React
 - iii. it has better community support
 - **VueJS:**
 - i. open source progressive JavaScript framework
 - ii. Any developer can easily understand and build interactive web interfaces in a matter of time

- 2 ways to start with react
 - React can be written in HTML directly: No much recommended
 - Using create-react-app utility: Steps to be followed
 - i. Install Node JS.
 - ii. Install create-react-app as global installation
 - iii. Change the directory to newly created application
 - iv. Run the application

- Basics
 - It embeds HTML into JavaScript code.
 - JSX is an HTML-like syntax used by React
 - This syntax is used by preprocessors (i.e., transpilers like babel) to transform HTML-like syntax into standard JavaScript objects that a JavaScript engine will parse.
 - Just like HTML, JSX tags have a tag name, attributes, and children.
- Advantages
 - It is faster than regular JavaScript
 - React uses components which put markup and logic together
 - Most of the errors can be found at compilation time.

- Rules for writing JSX:
 - Creating and rendering JSX using ReactDOM.render
 - With JSX you can write expressions inside curly braces { }.
 - To write HTML on multiple lines, put the HTML inside parentheses
 - The HTML code must be wrapped in ONE top level element
 - HTML elements must be properly closed